8

The Search

in which a method to search
for black holes in the sky
is proposed and pursued
and succeeds (probably)

The Method

Imagine yourself as J. Robert Oppenheimer. It is 1939; you have just convinced yourself that massive stars, when they die, must form black holes (Chapters 5 and 6). Do you now sit down with astronomers and plan a search of the sky for evidence that black holes truly exist? No, not at all. If you are Oppenheimer, then your interests are in fundamental physics; you may offer your ideas to astronomers, but your own attention is now fixed on the atomic nucleus—and on the outbreak of World War 11, which soon will embroil you in the development of the atomic bomb. And what of the astronomers; do they take up your idea? No, not at all. There is a conservatism abroad in the astronomical community, except for that "wild man" Zwicky, pushing his neutron stars (Chapter 5). The worldview that rejected Chandrasekhar's maximum mass for a white-dwarf star (Chapter 4) still holds sway.

Imagine yourself as John Archibald Wheeler. It is 1962; you are beginning to be convinced, after mighty resistance, that some massive stars must create black holes when they die (Chapters 6 and 7). Do you now sit down with astronomers and plan a search for them? No, not at all. If you are Wheeler, then your interest is riveted on the fiery marriage of general relativity with quantum mechanics, a marriage that may take place at the